Introduction to Classical Cryptography

Overview

- Introduction
- Simplified Model of Symmetric Encryption
- Parameters of Symmetric Encryption
- Cryptography
- Substitution Techniques
- Transposition Techniques
- Practice Problems
- Summary

Introduction … Contd.

Basic Terms

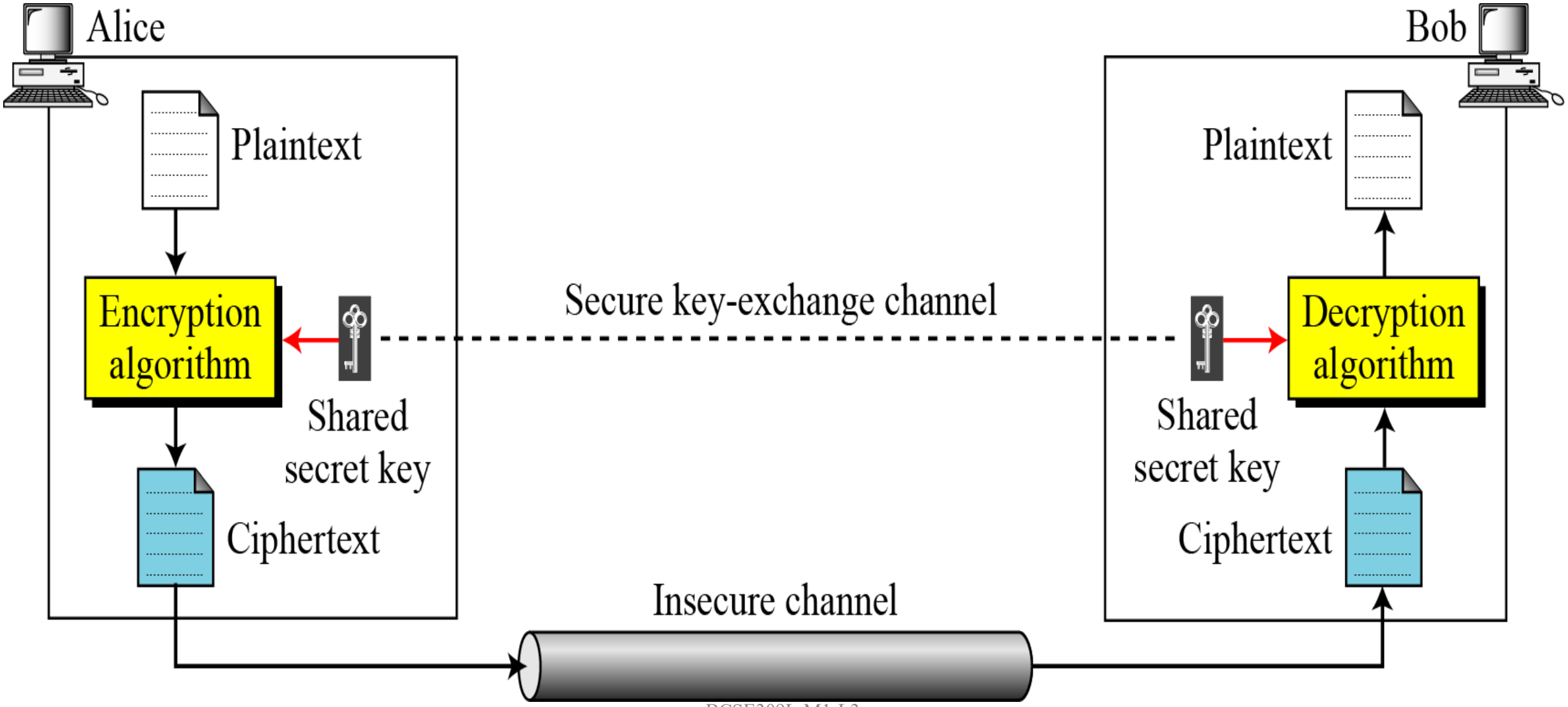
1) Plaintext – Original message
2) Ciphertext – Coded message
3) Enciphering/Encryption – Process of converting plaintext to ciphertext
4) Deciphering/Decryption – Restoring the plaintext from ciphertext
5) Cryptography – Schemes used for encryption constitute the area of study
6) Cryptographic System/Cipher – Scheme
7) Cryptanalysis – Techniques used for deciphering a message without any knowledge of the enciphering details
8) Cryptology – Areas of cryptography and cryptanalysis together

Introduction

Symmetric Encryption

- Conventional / private-key /single-key
- Sender and recipient share a common key
- All classical encryption algorithms are private-key
- Was only type prior to invention of public-key in 1970's
- And by far most widely used (still)
- Significantly faster than public-key cryptography

General idea of symmetric-key cipher



Model of Symmetric Cryptosystem … Contd.

Requirements for Secure Use of Conventional Encryption

- Strong encryption algorithm
- Secret key known only to sender / receiver

Mathematically,

- $Y = E(K, X) = E_K(X) = \{X\}_K$
- $X = D(K, Y) = D_K(Y)$

- Assume encryption algorithm is known
  - ***Kerchoff's Principle:*** *security in secrecy of key alone, not in obscurity of the encryption algorithm*
- Secure channel to distribute key
  - *Central problem in symmetric cryptography*

Cryptography

Characterizes cryptographic system by:

- Type of encryption operations used
  - Substitution
  - Transposition
- Number of keys used
  - Single-key or private
  - Two-key or public
- Way in which plaintext is processed
  - Block
  - Stream

Introduction to Substitution Techniques

Classical Substitution Ciphers

- Letters of plaintext are replaced by other letters or by numbers or symbols
- If plaintext is viewed as a sequence of bits, then substitution involves replacing plaintext bit patterns with ciphertext bit patterns

Types of Substitution Ciphers

- **Caesar Cipher (Shift Cipher)**
- **Affine Cipher**
- **Vignere Cipher**
- **Hill Cipher**
- **Playfair Cipher**
- **One-Time Pad**
- ~~Monoalphabetic Ciphers~~
- ~~Polyalphabetic Ciphers~~

1. Caesar Cipher

- Earliest known substitution cipher
- by Julius Caesar
- First attested use in military affairs
- Replaces each letter by kth letter on

Example (k=3)

Plaintext : meet me after the toga party

Ciphertext: PHHW PH DIWHU WKH WRJD SDUWB

1. Caesar Cipher … Contd.

- Can define transformation as:

  a b c d e f g h i j k l m n o p q r s t u v w x y z = IN

  D E F G H I J K L M N O P Q R S T U V W X Y Z A B C = OUT

- Mathematically give each letter a number

  a b c d e f g h i j k l m n o p q r s t u v w x y z

  0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

- Then have Caesar (rotation) cipher as:
  - *Ciphertext*, $c = E(k, p) = (p + k) \bmod (26)$
  - *Plaintext*, $p = D(k, c) = (c - k) \bmod (26)$

1. Cryptanalysis of Caesar Cipher

- Only have 26 possible ciphers
- A maps to A,B,..,Z
- Simply try each in turn
- Brute force search
  - Given ciphertext, just try all shifts of letters
  - Do need to recognize when have plaintext
  - Eg. break ciphertext "**GCUA VQ DTGCM**"

2. Affine Cipher

- Monoalphabetic Substitution cipher
- The whole process relies on working *modulo m*, i.e., the length of the alphabet used.
- By performing a calculation on the plaintext letters, we encipher the plaintext.

2. Affine Cipher Encryption

- Transform each of the letters in the plaintext alphabet to the corresponding integer in the range 0 to m-1.
- The encryption process for each letter is given by

$$\mathbf{E}(x) = (ax + b) \bmod m$$

  - where a and b are the key for the cipher.
- Multiply the integer value for the plaintext letter by a, and then add b to the result.
- Finally, take this modulus m (i.e., take the remainder when the solution is divided by m, or take away the length of the alphabet until

  get a number less than this length).

2. Affine Cipher Encryption … Contd.

- Encrypt the plaintext "affine cipher", using the key $a = 5$, $b = 8$
- Firstly find the integer value of each of the letters in the plaintext alphabet

Plaintext Alphabet	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
Plaintext Value	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

2. Affine Cipher Encryption … Contd.

- With the integer values of the plaintext letters found, the next step is to perform the calculations on those values.
- The calculation needed is (5x+8).
- Finally, ensure that all the answers are calculated mod 26 and convert the integers back to ciphertext letters.

Plaintext	a	f	f	i	n	e		c	i	p	h	e	r
x	0	5	5	8	13	4		2	8	15	7	4	17
5x+8	8	33	33	48	73	28		18	48	83	43	28	93
(5x+8) mod 26	8	7	7	22	21	2		18	22	5	17	2	15
Ciphertext	I	H	H	W	V	C		S	W	F	R	C	P

2. Affine Cipher Decryption

- Perform the opposite (or inverse) functions on the ciphertext to retrieve the plaintext.
- First step is to convert each of the ciphertext letters into their integer values.
- Perform the following calculation on each integer

$$\mathbf{D}(x) = c(x - b) \bmod m$$

  - where c is the modular multiplicative inverse of a.
- i.e., $a \times c = 1 \bmod m$ (c is the number such that when you multiply a by it, and keep taking away the length of the alphabet, you get to 1).

2. Affine Cipher Decryption … Contd.

- Decrypt the ciphertext "IHHWVC SWFRCP", using a key of $a = 5$, $b = 8$.
- The first step here is to find the inverse of a, which in this case is 21 (since 21 x 5 = 105 = 1 mod 26).
- Perform the inverse calculations on the integer values of the ciphertext.
- In this case the calculation in $21(y - 8)$.
- Once again, take these answers modulo 26, and finally convert the integers back to plaintext letters.

Ciphertext	I	H	H	W	V	C		S	W	F	R	C	P
y	8	7	7	22	21	2		18	22	5	17	2	15
21(y - 8)	0	-21	-21	294	273	-126		210	294	-63	189	-126	147
21(y - 8) mod 26	0	5	5	8	13	4		2	8	15	7	4	17
Plaintext	a	f	f	i	n	e		c	i	p	h	e	r

3. Vigenere Cipher

- Simplest substitution cipher
- Multiple Caesar ciphers
- Key is multiple letters long $K = k_1\ k_2\ \ldots\ k_d$
- i^{th} letter specifies i^{th} alphabet to use
- Use each alphabet in turn
- Repeat from start after d letters in message
- Decryption simply works in reverse
- Key stream does not depend on the plaintext characters
- Depends only on the position of the character in the plain text

Plain text: $P = P_1P_2P_3 \ldots$

Cipher text: $C = C_1C_2C_3 \ldots$

Key stream: $K = [(k_1, k_2, \ldots, k_m), (k_1, k_2, \ldots, k_m), \ldots]$

Encryption: $C_i = P_i + k_i$

Decryption: $P_i = C_i - k_i$

3. Vigenere Cipher … Contd.

- Encrypt the message **"She is listening"** using the 6-character keyword **"PASCAL"**. The initial key stream is **(15,0,18,2,0,11).** The key stream is the repetition of this initial key stream (as many times as needed)

Plaintext:	s	h	e	i	s	l	i	s	t	e	n	i	n	g
P's values:	18	07	04	08	18	11	08	18	19	04	13	08	13	06
Key stream:	***15***	***00***	***18***	***02***	***00***	***11***	***15***	***00***	***18***	***02***	***00***	***11***	***15***	***00***
C's values:	**07**	**07**	**22**	**10**	**18**	**22**	**23**	**18**	**11**	**6**	**13**	**19**	**02**	**06**
Ciphertext:	**H**	**H**	**W**	**K**	**S**	**W**	**X**	**S**	**L**	**G**	**N**	**T**	**C**	**G**

3. Vigenere Cipher … Contd.

Vigenere Tableau / Vigenere Table / Vigenere Square

- First row of this table has the 26 English letters
- Shows the plain text character to be encrypted
- Starting with the second row, each row has the letters shifted to the left one position in a cyclic way
- Eg., when **B** is shifted to the first position on the second row, the letter **A** moves to the end
- The first column contains the characters to be used by the key

3. Vigenere Cipher … Contd.

Vigenere Tableau / Vigenere Table / Vigenere Square

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

3. Vigenere Cipher … Contd.

Vigenere Tableau / Vigenere Table / Vigenere Square

To find the cipher text for the plaintext **"she is listening "** using the word **"PASCAL"** as the key

- Find **"s"** in the first row**,** **"p"** in the first column, the cross section is the cipher text character **"H"**
- Find **"h"** in the first row, **"A"** in the first column, the cross section is the cipher text character **"H"**
- And so on…..

3. Vigenère Cipher … Contd.

Security Analysis

- Have multiple ciphertext letters for each plaintext letter
- Hence letter frequencies are obscured
- But not totally lost
- Start with letter frequencies
  - See if it looks monoalphabetic or not
- If not, then need to determine number of alphabets, since then can attack each

4. Hill Cipher

- Invented by L. S. Hill in 1929.
- Inputs : String of English letters, A,B,…,Z.

  An n×n matrix **K**, with entries drawn from 0,1,…,25.

  (The matrix **K** serves as the secret key.)
- Divide the input string into blocks of size n.
- Identify A=0, B=1, C=2, …, Z=25.
- Encryption: Multiply each block by **K** and then reduce mod 26.
- Decryption: multiply each block by the inverse of **K**, and reduce mod 26.

4. Hill Cipher … Contd.

- The decryption must be the inverse function of the encryption function.
  - It is required that $\mathbf{K}^{-1}\ \mathbf{K} = \mathbf{I}_n \bmod 26$.

$$K^{-1} = \frac{1}{\det K} Adj(K)$$

- Provided that det(**K**) has a multiplicative inverse mod 26, i.e., if det(**K**) and n has no common factor, the inverse of **K** can be computed by the adjoint formula for matrix inverse.
- Inverse of an integer mod 26 can be obtained by trial and error/Extended Euclidean Algorithm.

4. Hill Cipher - Example

Encryption

- Plain text: "LOVE", Secret Key: $\begin{bmatrix} 20 & 3 \\ 15 & 7 \end{bmatrix}$
- "LO" → $\begin{bmatrix} 20 & 3 \\ 15 & 7 \end{bmatrix} \begin{bmatrix} 11 \\ 14 \end{bmatrix} = \begin{bmatrix} 262 \\ 263 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \mod 26$
- "VE" → $\begin{bmatrix} 20 & 3 \\ 15 & 7 \end{bmatrix} \begin{bmatrix} 21 \\ 4 \end{bmatrix} = \begin{bmatrix} 432 \\ 343 \end{bmatrix} = \begin{bmatrix} 16 \\ 5 \end{bmatrix} \mod 26$
- 2, 3, 16, 5 are transformed to cipher text "CDQF"

A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12

N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25

4. Hill Cipher - Example … Contd.

How to decode?

- Given "CDQF", and the encryption matrix $\begin{bmatrix} 20 & 3 \\ 15 & 7 \end{bmatrix}$
- How do we decrypt?
  - We need to compute the inverse of $\begin{bmatrix} 20 & 3 \\ 15 & 7 \end{bmatrix}$

- Remind that all arithmetic are mod 26.
- There is no fraction and care should be taken in computing multiplicative inverse mod 26.

4. Hill Cipher - Example … Contd.

Determinant

- The determinant of $\begin{bmatrix} 20 & 3 \\ 15 & 7 \end{bmatrix}$ equals 20(7)-3(15)=95, which is 17 mod 26.
- Find the multiplicative inverse of 17 mod 26, i.e., find integer b such that 17x b= 1 mod 26.
- Just try all 26 possibilities for x:

17×1 = 17 mod 26	17×8 = 6 mod 26	17×15 = 21 mod 26	17×22 = 10 mod 26
17×2= 8 mod 26	17×9= 23 mod 26	17×16= 12 mod 26	17×23= 1 mod 26
17×3 = 25 mod 26	17×10 = 14 mod 26	17×17 = 3 mod 26	17×24 = 18 mod 26
17×4 = 16 mod 26	17×11 = 5 mod 26	17×18 = 20 mod 26	17×25 = 9 mod 26
17×5 = 7 mod 26	17×12 = 22 mod 26	17×19 = 11 mod 26	17×0 = 0 mod 26
17×6 = 24 mod 26	17×13 = 13 mod 26	17×20 = 2 mod 26	
17×7 = 15 mod 26	17×14 = 4 mod 26	17×21 = 19 mod 26	

4. Hill Cipher - Example … Contd.

Computing the inverse of mod 26

- From 17×23= 1 mod 26, we know that the multiplicative inverse of 17 mod 26 is 23.
- Using the formula for 2 × 2 matrix inverse

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = (ad - bc)^{-1} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

we get

Replace $(17)^{-1}$ mod 26 by 23

$$\begin{bmatrix} 20 & 3 \\ 15 & 7 \end{bmatrix}^{-1} = (17)^{-1} \begin{bmatrix} 7 & -3 \\ -15 & 20 \end{bmatrix} = 23 \begin{bmatrix} 7 & 23 \\ 11 & 20 \end{bmatrix} = \begin{bmatrix} 5 & 9 \\ 19 & 18 \end{bmatrix} \text{ mod } 26$$

4. Hill Cipher - Example … Contd.

Decryption

- Given the ciphertext "CDQF", we decrypt by multiplying by $\begin{bmatrix} 5 & 9 \\ 19 & 18 \end{bmatrix}$

$$\begin{bmatrix} 5 & 9 \\ 19 & 18 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 37 \\ 92 \end{bmatrix} = \begin{bmatrix} 11 \\ 14 \end{bmatrix} \mod 26$$

$$\begin{bmatrix} 5 & 9 \\ 19 & 18 \end{bmatrix} \begin{bmatrix} 16 \\ 5 \end{bmatrix} = \begin{bmatrix} 125 \\ 394 \end{bmatrix} = \begin{bmatrix} 21 \\ 4 \end{bmatrix} \mod 26$$

A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12

N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25

- From the table, 11, 14, 21, 4 is "LOVE".

5. Playfair Cipher

- Not even the large number of keys in a monoalphabetic cipher provides security
- One approach to improving security was to encrypt multiple letters
- The **Playfair Cipher** is an example
- Invented by Charles Wheatstone in 1854, but named after his friend Baron Playfair

5. Playfair Cipher … Contd.

Playfair Matrix

- A 5X5 matrix of letters based on a keyword
- Fill in letters of keyword (minus duplicates)
- Fill rest of matrix with other letters
- E.g., keyword MONARCHY

M	**O**	**N**	**A**	**R**
C	**H**	**Y**	**B**	**D**
E	**F**	**G**	**I/J**	**K**
L	**P**	**Q**	**S**	**T**
U	**V**	**W**	**X**	**Z**

5. Playfair Cipher … Contd.

Encryption and Decryption

- Plaintext is encrypted two letters at a time

1) If a pair is a repeated letter, insert filler like 'X'
2) If both letters fall in the same row, replace each with letter to right (wrapping back to start from end)
3) If both letters fall in the same column, replace each with the letter below it (wrapping to top from bottom)
4) Otherwise each letter is replaced by the letter in the same row and in the column of the other letter of the pair

- Ciphertext is decrypted two letters at a time with the reverse process of encryption

5. Playfair Cipher … Contd.

Example-1

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

- Message = Move forward
- Plaintext = mo ve fo rw ar d**x**
- Here x is just a filler, message is padded and segmented
- mo → ON; ve → UF; fo →PH; rw → NZ; ar → RM; dx →BZ
- Ciphertext = ON UF PH NZ RM BZ

5. Playfair Cipher … Contd.

Example-2

- Message = attack

at	ta	ck
RS	SR	DE

- Ciphertext = RSSRDE

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

5. Playfair Cipher … Contd.

Example-3

- Message = mosque

mo	sq	ue
ON	TS	ML

- Ciphertext = ONTSML

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

5. Playfair Cipher … Contd.

Example-4

- Message = Hide the gold under the carpet
- Key: Neso Academy

N	E	S	O	A
C	D	M	Y	B
F	G	H	I/J	K
L	P	Q	R	T
U	V	W	X	Z

- Plaintext:
- Ciphertext:

Hi	de	th	eg	ol	du	nd	er	th	ec	ar	pe	tx
IK	GD	QK	DP	NR	CV	EC	OP	QK	ND	OT	VD	RZ

5. Playfair Cipher … Contd.

Security

- Since have 26 x 26 = 676 digrams
- Would need a 676 entry frequency table to analyse (versus 26 for a monoalphabetic)
- And correspondingly more ciphertext
- Was widely used for many years
- E.g., by US & British military in WW1&2
- It can be broken, given a few hundred letters
- Since still has much of plaintext structure

6. One-Time Pad

- If a truly random key as long as the message is used, the cipher will be secure is called a One-Time Pad (OTP)
- Is unbreakable since ciphertext bears no statistical relationship to the plaintext
- Since for **any plaintext** & **any ciphertext** there exists a key mapping one to other
- Can only use the key **once** though
- **Problems in generation & safe distribution of key**

Introduction to Transposition Ciphers

- Now consider classical **transposition** or **permutation** ciphers
- These hide the message by rearranging the letter order
- Without altering the actual letters used
- Can recognize these since have the same frequency distribution as the original text

7. Rail Fence Cipher

- Write message letters out diagonally over a number of rows
- Then read off cipher row by row
- Eg., Plaintext: meet me after the toga party
- Write message out as:
  - m e m a t r h t g p r y
  - e t e f e t e o a a t

- Ciphertext: MEMATRHTGPRYETEFETEOAAT

8. Row Transposition Cipher

- More complex transposition
- Write letters of message out in rows over a specified number of columns
- Then reorder the columns according to some key before reading off the rows
- Eg., Plaintext: attack postponed until two am
  - Key: 4312567
  - Column Out 4 3 1 2 5 6 7
  - Plaintext:

a	t	t	a	c	k	p
o	s	t	p	o	n	e
d	u	n	t	i	l	t
w	o	a	m	x	y	z

Ciphertext: TTNAAPTMTSUOAODWCOIXKNLYPETZ

9. Block Transposition Cipher

- Arbitrary block transposition may be used
- Specify permutation on block
- Repeat for each block of plaintext
  - Key: 4931285607
  - Plaintext: attackpost poneduntil twoamxyzab

    0123456789

  - Ciphertext: CTATTSKPAO DLEONIDUPT MBAWOAXYTZ

Practice Problem 1– Caesar Cipher

Plaintext : HELLO
Ciphertext : EBIIL

Key - ?

Plaintext : ATTACKATONCE

Ciphertext : EXXEGOEXSRGI

Key - ?

Practice Problem 2 – Playfair Cipher

1) Construct a playfair matrix with the key "decryption".
2) Construct a playfair matrix with the key "largest".
3) Using the playfair matrices constructed in (1) and (2), encrypt the following message:

   "I only regret that I have but one life to give for my country"

Practice Problem 3 – Affine Cipher

1) Encrypt the plaintext "cryptography" using the key $a = 2$ and $b = 7$.
2) Encrypt the plaintext "continuous assessment test" using the key $a = 6$ and $b = 9$.
3) Decrypt the cipher text obtained in (1) and (2) using Affine Cipher.

Practice Problem 4 – Vignere Cipher

1) Encrypt the plaintext, "We are discovered save yourself" using the keyword, *deceptive* using Vigenere Cipher.
2) Encrypt the plaintext, "To be or not to be that is the question" using the keyword, *relations* using Vigenere Tableau.
3) Decrypt the ciphertexts obtained in (1) and (2) using Vigenere Cipher

Practice Problem 5 – Hill Cipher

1) Encrypt the plaintext "paymoremoney" by using the key $K=\begin{bmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{bmatrix}$
2) Decrypt the ciphertext obtained in (1).
3) Encrypt the plaintext "act" by using the key $K=\begin{bmatrix} 6 & 24 & 1 \\ 13 & 16 & 10 \\ 20 & 17 & 15 \end{bmatrix}$
4) Decrypt the ciphertext obtained in (3).

Summary

Discussed about

- Introduction
- Simplified Model of Symmetric Encryption
- Ingredients of Symmetric Encryption
- Cryptography
- Substitution Techniques
- Transposition Techniques
- Practice Problems

Extra

Monoalphabetic Ciphers

- Rather than just shifting the alphabet
- Could shuffle (permute) the letters arbitrarily
- Each plaintext letter maps to a different random ciphertext letter
- Hence key is 26 letters long

Plain	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
Cipher	D	K	V	Q	F	I	B	J	W	P	E	S	C	X	H	T	M	Y	A	U	O	L	R	G	Z	N

Plaintext: ifwewishtoreplaceletters

Ciphertext: WIRFRWAJUHYFTSDVFSFUUFYA

Monoalphabetic Ciphers … Contd.

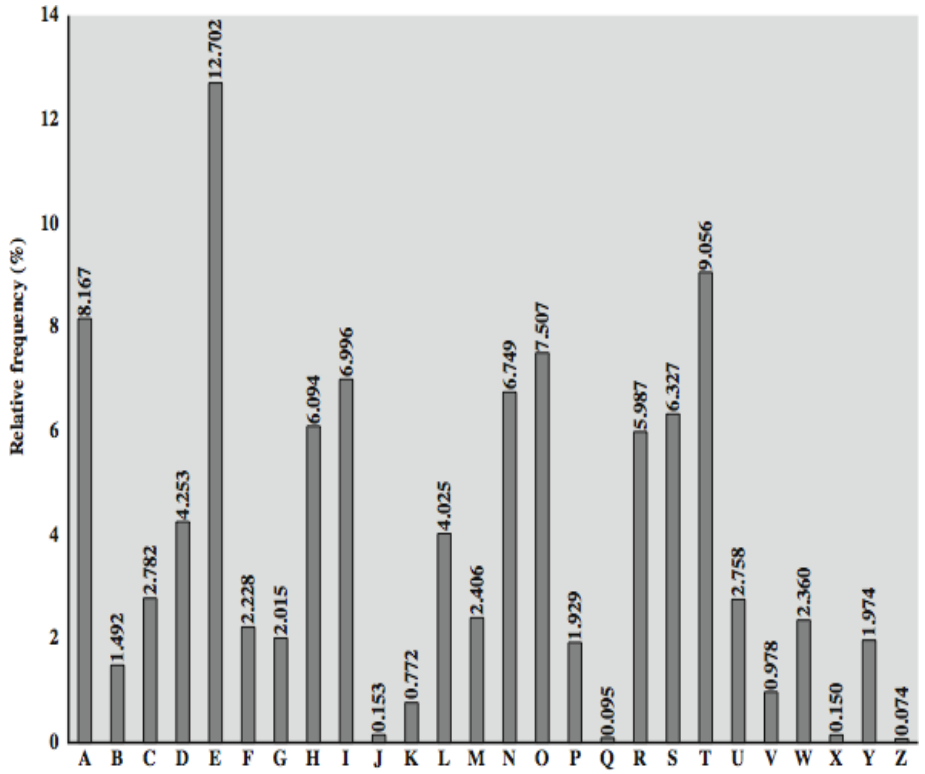
Security

- Key size is now 26 characters…
- Now have a total of $26! = 4 \times 10^{26}$ keys
- With so many keys, might think is secure
- But would be **!!!WRONG!!!**
- Problem is language characteristics

Monoalphabetic Ciphers … Contd.

Language Redundancy and Cryptanalysis

- Human languages are redundant
- E.g., "th lrd s m sh ph rd sh ll nt "
- Letters are not equally commonly used
- In English E is by far the most common letter
  - Followed by T,R,N,I,O,A,S
- Other letters like Z,J,K,Q,X are fairly rare
- Various languages have tables of single, double, & triple letter frequencies



Monoalphabetic Ciphers … Contd.

Use in Cryptanalysis

- Key concept - monoalphabetic substitution ciphers do not change relative letter frequencies
- Discovered by Arabian scientists in 9th century
- Calculate letter frequencies for ciphertext
- Compare counts/plots against known values
- If Caesar cipher look for common peaks/troughs
  - Peaks at: A-E-I triple, N-O pair, R-S-T triple
  - Troughs at: J-K, U-V-W-X-Y-Z
- For monoalphabetic must identify each letter
  - Tables of common double/triple letters help
  - Digrams and Trigrams
- Amount of ciphertext is important – statistics!

Monoalphabetic Ciphers … Contd.

Example - Cryptanalysis

- Given ciphertext:
  UZQSOVUOHXMOPVGPOZPEVSGZWSZOPFPESXUDBMETSXAIZ
  VUEPHZHMDZSHZOWSFPAPPDTSVPQUZWYMXUZUHSX
  EPYEPOPDZSZUFPOMBZWPFUPZHMDJUDTMOHMQ
- Guess P & Z are e and t
- Guess ZW is th and hence ZWP is “the”
- Proceeding with trial and error finally get:
  it was disclosed yesterday that several informal but direct contacts have been made
  with political representatives of the viet cong in moscow

Polyalphabetic Substitution Ciphers

- Relationship between a character in the plaintext to a character in the cipher text is one-to-many
- Improve security using multiple cipher alphabets
- Make cryptanalysis harder with more alphabets to guess and flatter frequency distribution
- Use a key to select which alphabet is used for each letter of the message
- Use each alphabet in turn
- Repeat from start after end of key is reached
- E.g., Vigenere Cipher and Vernam Cipher